Does Cough Audio Add Value Under Country Shift?

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Abstract

Smartphone cough recordings have been proposed as a cheap, specimen-free way to screen for tuberculosis (TB), yet models that perform well in one study often fail in another, and it is unclear whether cough audio adds anything to the questions a health worker already asks. We study this by holding out one country at a time in the CODA TB DREAM data and, while access is finalised, have built and validated the full pipeline on the Zambia CIDRZ dataset, where every participant was recorded by four devices at once. The pipeline reproduces published baselines, and on the same 325 participants a cough-only model and a questionnaire-only model perform equally (AUROC 0.740 and 0.749). An analysis of the cough features shows why transfer is hard: a simple classifier recognises the recording phone almost perfectly (AUROC 0.99) but TB status only weakly (0.67). Recording conditions, which differ between countries, dominate what the features encode.

1 Motivation and objectives

Tuberculosis is the leading infectious cause of death worldwide, and confirming it usually requires sputum and a laboratory. The World Health Organization has therefore prioritised a non-sputum *triage* test that decides who should be sent for confirmation, with at least 90% sensitivity and 70% specificity [1]. The CODA TB DREAM Challenge reported AUROC 0.78 to 0.83 for cough combined with routine clinical variables [2], but such results do not necessarily transfer: external validation in Peru reported AUROC of only 0.48 to 0.62 [3], and a cross-dataset study found that learned cough representations cluster by recording device rather than by TB status, while a clinical baseline transfers more consistently [4]. The clinics that would benefit most never contributed training data, which is precisely where current models are least reliable.

Objective. We ask whether self-supervised cough representations (HeAR and OPERA) improve TB triage beyond routine clinical variables when an entire country is excluded from model development, and whether adapting the audio encoder improves or reduces performance on the held-out country. The quantity of interest is the paired gain $\Delta = \mathrm{AUROC}(\text{questionnaire} + \text{cough}) - \mathrm{AUROC}(\text{questionnaire})$ on the same held-out patients.

Progress to date. Access to CODA is being finalised. In the meantime we have developed the full pipeline on the Zambia CIDRZ dataset, which serves the country-shift study in three ways: it validates the pipeline against published results; because the same coughs were recorded on several devices at two clinics, it isolates two ingredients that country holdout mixes together, device fully and clinic in part; and since Zambia is not a CODA country, it will serve as an additional, fully external test country.

33rd Conference on Neural Information Processing Systems (NeurIPS 2019), Vancouver, Canada.

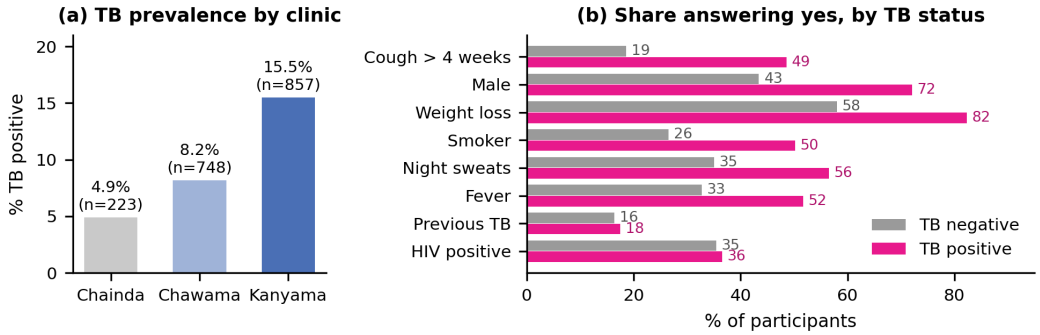


Figure 1: Exploratory analysis of the 1,828 Zambia participants: (a) TB prevalence by clinic; (b) share answering yes, sorted by the difference.

2 Related work and background

Zhang et al. [4] are our closest reference. Evaluating CODA, TBscreen [5] and Zambia, they report within-Zambia AUROC of 0.755 ± 0.056 for a mel-spectrogram VGGish model [6] and 0.767 ± 0.025 for logistic regression on clinical variables, and they suggest, without testing it, that clinical variables could provide a stable baseline to which audio adds complementary information. HeAR, pretrained on 313 million unlabelled body-sound clips [7], and OPERA, an open respiratory encoder never benchmarked on TB [8], are our candidate encoders, and LoRA [9] adapts them cheaply. Coppock et al. [10] showed that COVID-19 cough classifiers fell from AUROC 0.85 to 0.62 once confounders were matched, and we follow the participant-disjoint cross-validation of [11]. HeAR has been combined with CODA clinical data [12], but with clip-level predictions on a balanced subset. We instead measure the additional value of audio for the same held-out patients in each country, reporting AUROC for direct comparison.

3 Methodology

3.1 Data

CODA TB DREAM [13] is the main study dataset: adults with cough enrolled consecutively at clinics in seven countries, each giving at least three solicited coughs on an Android phone, with Xpert MTB/RIF Ultra and culture as the reference standard. The accessible partition holds about 1,100 participants, and our Intended Data Use statement has been submitted.

Zambia CIDRZ [7] is our development set and external test country. It holds questionnaire and laboratory data for 1,828 adults from three Lusaka clinics and 1,481 sessions in which each participant gave three single coughs and a burst of several, recorded simultaneously by an audio recorder and three phones: Phone A (Pixel 3a, low tier), Phone B (Galaxy A12) and Phone C (Galaxy A22, high tier).

Exploratory analysis. TB prevalence differs sharply by clinic, from 4.9% at Chainda-South to 15.5% at Kanyama (Figure 1a), and is higher still among participants with audio (about 9% at Chawama and 22% at Kanyama). Several questionnaire answers separate TB-positive from TB-negative participants clearly (Figure 1b): coughing for more than four weeks (49% against 19%), being male (72% against 43%) and weight loss (82% against 58%), whereas HIV status and previous TB barely differ. The questionnaire is therefore a strong opponent: cough audio must carry information beyond "has been coughing for weeks" to be worth adding. The recordings also differ visibly by device (Figure 2).

Preparation. We applied the 62 published filename corrections; audited every recording by cross-correlating each participant's files, since simultaneous recordings of one session line up closely while different people's do not, which confirmed 357 of 363 participants and removed seven foreign or unmatched files; cut sessions into coughs with Zhang et al.'s CLAP-based extractor [15], which cuts

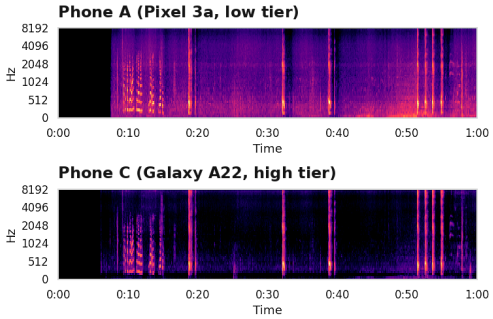


Figure 2: One session on two phones (time against pitch). The coughs line up, but Phone A records low-pitched background noise that Phone C does not.

every device at identical times; and kept the Chawama and Kanyama clinics and the three phones, giving 325 participants (57 TB-positive) and 5,287 clips. CODA ships pre-cut clips, so only the evaluation steps carry over to it.

3.2 Models, metric and loss

The arms follow the proposal: Model B (questionnaire only), arm A1 (MFCC network), arms A2 and A3 (frozen HeAR and OPERA) and arm A4 (HeAR with LoRA), plus an audio-only ablation. To date, Model B and the audio-only arm are complete, A2 and A3 are next, and A1 and A4 follow on CODA. **Model B** is a regularised logistic regression on nine questionnaire answers. The **audio-only arm** is VGGish [6], pretrained on AudioSet and fine-tuned on 0.96-second log-mel spectrograms, with a participant's score taken as the median over their clips. We used VGGish first, rather than the proposed MFCC network, because a published Zambia result exists for it and so provides a reproduction check. **Model A** will combine the questionnaire with the audio representation, either through a two-layer MLP or by adding a single out-of-fold audio score to the logistic regression, so that a zero audio weight recovers Model B. Audio models minimise class-weighted binary cross-entropy.

AUROC is the probability that a randomly chosen TB patient receives a higher score than a randomly chosen non-TB patient, so 0.5 is chance. **Specificity at 90% sensitivity** is the share of non-TB patients spared a laboratory test when a model catches nine of every ten TB cases. Participants, never individual clips, are split into five folds stratified by site and TB status and repeated twice, every model uses the same saved folds, and paired differences are tested with DeLong's method [14]. On CODA the same protocol will hold out one country at a time, with all fitting done on the remaining countries.

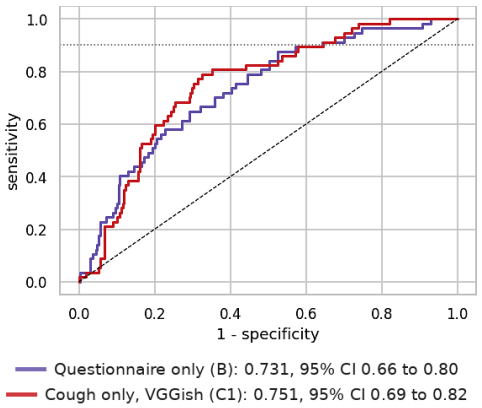
4 Baseline and experimental iterations

Baseline reproduction. Both baselines land within one standard deviation of the published values (Figure 3b), and all numbers come from our own runs. The ROC curves of the two models nearly overlap (Figure 3a).

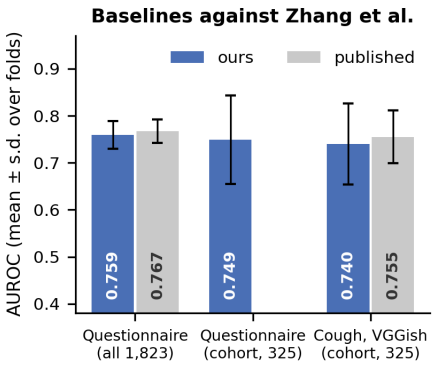
Iterations. We ran the full pipeline twice. Between runs we corrected three defects in the published training code: pretrained weights were copied by tensor shape, which left four VGGish bias vectors random and gave four others another layer's bias; the best training epoch was never restored; and test clips were augmented and duplicated. We also replaced "keep the first file found" with the audit. These changes altered a dozen files but reshuffled every fold, which proved informative.

5 Results and analysis

Cough alone ties with the questionnaire. On the same participants the paired difference between the cough and questionnaire models is $+0.020$ (DeLong 95% CI -0.069 to 0.109) in one repeat and -0.004 (-0.094 to 0.086) in the other. At 90% sensitivity both reach only 35 to 36% specificity, so either model would still send about 64 of every 100 non-TB patients for a laboratory test. The cough



(a) ROC curves, pooled test predictions



(b) Our baselines against the published values

Figure 3: Baselines on Zambia: the two models are indistinguishable and both reproduce Zhang et al. [4]. Dotted line: 90% sensitivity.

Table 1: What a linear classifier recovers from the cough features, tested on unseen participants. Chance is 0.5 for AUROC and 0.33 for phone accuracy.

	Phone		TB status (AUROC)	
Features	Accuracy	AUROC	per clip	per person
Spectrogram summary	0.955	0.994	0.642	0.665
VGGish embedding (pretrained)	0.772	0.912	0.601	0.651

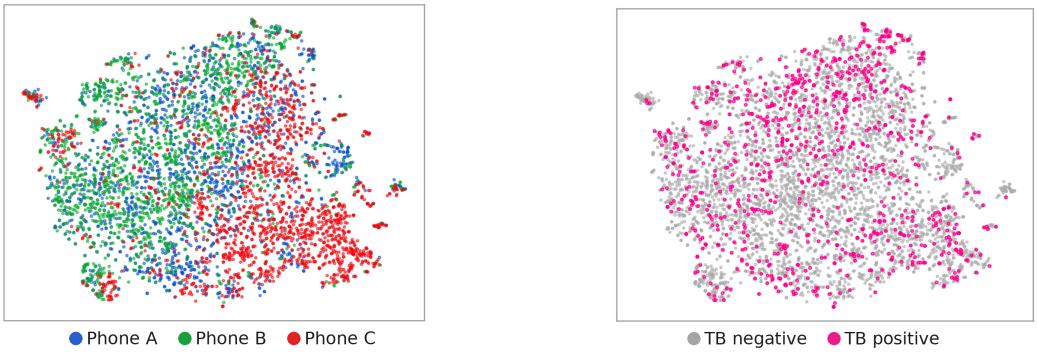
carries some TB information, yet no more than the questionnaire, which raises the question of what else its features carry.

Recording conditions dominate the features. We represented every clip as a simple spectrogram summary (mean and spread of each mel band) and as the pretrained VGGish embedding, and asked a linear classifier, always tested on unseen participants, to recover the phone or the TB status (Table 1). The phone is recovered almost perfectly from the spectrogram summary (AUROC 0.99, accuracy 95% against 33% chance) and well from VGGish (0.91), whereas TB status is recovered only weakly (0.65 to 0.67 per person). A two-dimensional map of the embeddings tells the same story (Figure 4): Phone C occupies its own region, Phones A and B overlap, and TB-positive clips are scattered across the whole map. The effect reaches predictions too, since the fine-tuned model scores the same participants differently on each phone, by 0.075 to 0.105 on average.

The weak TB signal is genuine. Kanyama's TB rate is more than twice Chawama's and the features partly reveal the clinic (AUROC 0.73), so a model could seem to detect TB by recognising Kanyama. Within Kanyama alone (47 TB-positive participants), where that shortcut gives no advantage, TB AUROC stays at 0.65 to 0.69, so the signal is real although weak.

Sensitivity to the data split. Our two runs differ mainly in fold assignment, yet the cough model moved from 0.761 to 0.740 and an apparent triage advantage for cough (42% against 24% specificity) disappeared (35% against 36%). With 57 TB-positive participants, differences below about 0.05 AUROC cannot be detected, and Chawama, with only 10 TB-positive participants, is too small to analyse on its own.

Implications for country shift. Together these results describe the problem our study targets. Cough audio holds a real but weak TB signal beside a much stronger signal from the recording conditions. Within one dataset a model tolerates that imbalance, but countries differ in phones, clinics and recording environments, so under country holdout the strong signal shifts while the weak one must carry the prediction. This is a plausible mechanism behind drops such as the Peruvian result, and the reason the gain must be measured on unseen countries.



(a) Coloured by phone

(b) The same map, coloured by TB status

Figure 4: t-SNE map of the pretrained VGGish embeddings of all 5,287 clips: phones form regions, while TB-positive clips are spread throughout.

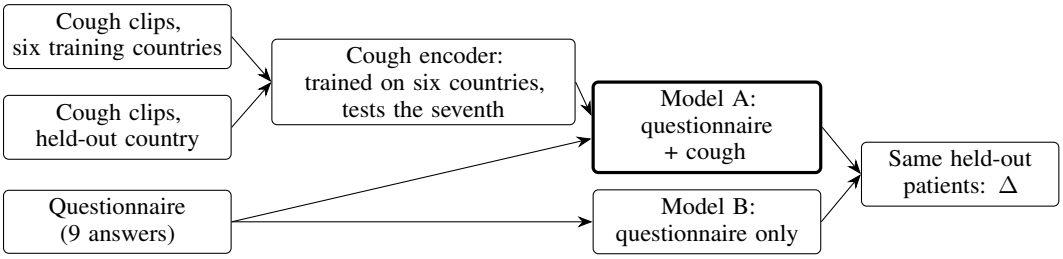


Figure 5: Leave-one-country-out design on CODA, with Zambia as an external eighth country. Model B sees no audio, so changes in Δ between countries reflect the cough pathway.

6 Extensions to be conducted

The next stage runs the study as proposed (Figure 5). On CODA, each country is held out in turn while models are developed on the remaining six, and Δ is measured per held-out country together with the mean and worst-country AUROC and specificity at a threshold fixed on the training countries. Arms A1 to A3 compare the MFCC network with frozen HeAR and OPERA, arm A4 tests whether LoRA adaptation to six countries helps or harms the seventh, and an attention-pooling ablation follows. Models developed on CODA will then be tested on Zambia as an external eighth country, using the nine clinical variables both datasets share [4].

7 Discussion: challenges and risks

Data access. CODA is controlled; our Intended Data Use statement is submitted and a faculty sign-off is being requested, and until access is granted the country-holdout experiments cannot start. **Compute.** The 395 GB Kenyan TBscreen dataset [5] exceeds our storage and GPU budget, which rules it out as a further external test. **Sample size.** If per-country positive counts in CODA are too small, we will merge the smallest countries, as planned; on Zambia, 57 TB-positive participants already make intervals wide. **Attribution.** Country holdout mixes device, clinic, recruitment and severity; the Zambia analyses isolate the first two only partly.

8 Future directions

Our next three actions are: (1) build the fusion model A and the HeAR and OPERA embeddings on Zambia now, so they run on CODA unchanged; (2) on receiving CODA, check per-country class balance and run arms B and A1 to A3 under random and leave-one-country-out splits; and (3) test the CODA models on Zambia.

Team and code. Data and audit; encoders; fusion; evaluation. `github.com/<org>/<repo>`

References

[1] World Health Organization, *High-Priority Target Product Profiles for New Tuberculosis Diagnostics: Report of a Consensus Meeting*. Geneva, Switzerland: WHO, 2014.

[2] D. Jaganath, S. K. Sieberts, M. Raberahona, *et al.*, "Accelerating cough-based algorithms for pulmonary tuberculosis screening: Results from the CODA TB DREAM Challenge," *Open Forum Infectious Diseases*, vol. 12, no. 10, p. ofaf572, 2025.

[3] A. J. Zimmer *et al.*, "External validation of cough-based algorithms for pulmonary tuberculosis screening from the CODA TB DREAM Challenge using cough data from Peru," *Scientific Reports*, 2026.

[4] W. Zhang, T. Teijeiro, J. Thevenot, and D. Atienza, "Why ML-based cough models do not generalize: A systematic cross-dataset evaluation for tuberculosis screening," *arXiv preprint arXiv:2608.25846*, 2026.

[5] M. Sharma, V. Nduba, L. N. Njagi, *et al.*, "TBscreen: A passive cough classifier for tuberculosis screening with a controlled dataset," *Science Advances*, vol. 10, no. 1, p. eadi0282, 2024.

[6] S. Hershey, S. Chaudhuri, D. P. W. Ellis, *et al.*, "CNN architectures for large-scale audio classification," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2017, pp. 131–135.

[7] S. Baur, Z. Nabulsi, W.-H. Weng, *et al.*, "HeAR: Health acoustic representations," *arXiv preprint arXiv:2403.02522*, 2024.

[8] Y. Zhang, T. Xia, J. Han, *et al.*, "Towards open respiratory acoustic foundation models: Pretraining and benchmarking," in *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track*, 2024.

[9] E. J. Hu, Y. Shen, P. Wallis, *et al.*, "LoRA: Low-rank adaptation of large language models," in *Proc. Int. Conf. Learning Representations (ICLR)*, 2022.

[10] H. Coppock, G. Nicholson, I. Kiskin, *et al.*, "Audio-based AI classifiers show no evidence of improved COVID-19 screening over simple symptoms checkers," *Nature Machine Intelligence*, vol. 6, no. 2, pp. 229–242, 2024.

[11] G. P. Kafentzis and E. Selisios, "Tuberculosis screening from cough audio: Baseline models, clinical variables, and uncertainty quantification," *Sensors*, vol. 26, no. 4, p. 1223, 2026.

[12] R. A. Buyung and W. Nugroho, "Multimodal deep learning for tuberculosis detection using cough audio and clinical data with health acoustic representations (HeAR)," *Int. J. Advanced Computer Science and Applications*, vol. 16, no. 10, 2025.

[13] S. Huddart, V. Yadav, S. K. Sieberts, *et al.*, "Solicited cough sound analysis for tuberculosis triage testing: The CODA TB DREAM Challenge dataset," *Scientific Data*, vol. 11, p. 1149, 2024.

[14] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[15] B. Elizalde, S. Deshmukh, and H. Wang, "Natural language supervision for general-purpose audio representations," in *Proc. IEEE Int. Conf. Acoustics, Speech and Signal Processing (ICASSP)*, 2024.